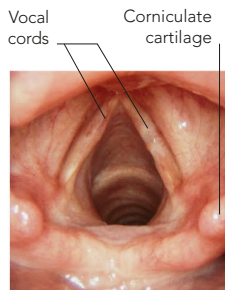


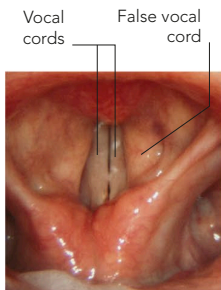
VOCALIZATION

The vocal cords are two bands of fibrous tissue within the larynx. During breathing they are separated by a V-shaped gap (the glottis). Sound is produced when the cords tighten together and vibrate as air from the lungs passes between them. Pitch varies according to the tension in the cords. The false vocal cords above help to close off the larynx during swallowing.



CORDS APART

A laryngoscope view shows the vocal cords during normal breathing, when air passes through the gap between them.

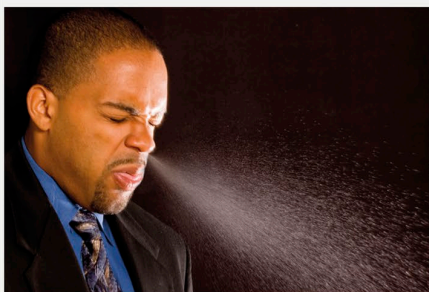


CORDS ADJACENT

Laryngeal muscles swing the arytenoid cartilages, to which the vocal cords are attached, and bring them together.

RESPIRATORY REFLEXES

The respiratory reflexes of coughing and sneezing aim to blow out excess mucus, dust, irritants, and obstructions – coughing from the lower pharynx, larynx, trachea, and lung airways, and sneezing from the nasal chambers and nasopharynx. For a cough, the lower pharynx, epiglottis, and larynx close so that air pressure builds up in the lungs, and is released explosively, rattling the vocal cords. In a sneeze, the tongue closes off the mouth, to force air up and out through the nose.



MUCUS SPRAY

Coughs (as shown here) and sneezes propel a spray of tiny mucus droplets from the respiratory airways for distances of up to 3m (10ft).

THE LARYNX

The larynx is sited between the pharynx and the trachea. It has a framework of nine cartilages, comprising the paired arytenoids, cuneiforms, and corniculates, and the unpaired epiglottic, thyroid, and cricoid. The thyroid cartilage forms a prominent mound under the skin of the neck, called the “Adam’s apple”, which is larger and more pronounced in adult males. The cartilages are held in position by numerous muscles and ligaments.

INTERNAL STRUCTURE

The larynx forms a hollow chamber through which air flows silently during normal breathing. The areas of cartilage tilt to bring the vocal cords together for speech.

